WEAKLY ETALE MORPHISMS ARE STABLE UNDER BASE CHANGE, VIA FLATNESS OF THE BASE-CHANGED DIAGONAL

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 19001-19010 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 19012-19018, the complete author proof (5 body lines).

Editorial note. Navigation stubs: 1 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Sixth member of the weakly etale series (L16089-L16091, L16119, L16146); the base-change law completing the toolkit.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Morphisms, lemma base-change-flat; base-change stability of unramified morphisms (Morphisms chapter); definition of weakly etale.

Source reference. Original lemma 01U9: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 19018–19018. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $X \to Y$ and $Y' \to Y$ be morphisms of schemes and let $X' = Y' \times_Y X$ be the base change of X.*

(1) *If $X \to X \times_Y X$ is flat, then $X' \to X' \times_{Y'} X'$ is flat.*
(2) *If $X \to Y$ is weakly étale, then $X' \to Y'$ is weakly étale.*

Proof. Assume $X \to X \times_Y X$ is flat. The morphism $X' \to X' \times_{Y'} X'$ is the base change of $X \to X \times_Y X$ by $Y' \to Y$. Hence it is flat by Morphisms, Lemmas 01U9. This proves (1). Part (2) follows from (1) and the fact (just used) that the base change of a flat morphism is flat. □

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.